Test Case Document

VendorSetu

Street Vendor Formalization & Micro-Credit Platform

Version 1.0 · Based on SRS v1.0 and Test Case Template (Doc T_01)

Prepared by: Nancy Yadav, <Member 2>, <Member 3>, <Member 4>

B.Tech CSE, Batch 2024 - CSE3005 Software Engineering - BML Munjal University

Submitted to: <Faculty name>

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1. Test Plan Summary

Item	Details
Objective	Verify that the four implemented modules of VendorSetu meet their functional requirements (SRS Section 4.1-4.6) and reject invalid input with the correct message.
Modules under test	1. Registration & Login 2. Vendor Profile & Daily Sales 3. Trust Scoring Engine 4. Loan Application & Bank Review 5. EMI Schedule & Repayment (Module 4 continued)
Test levels	System testing through the web interface: 29 cases - the tester fills forms and clicks buttons; the output is the message or value shown on screen. Unit testing of the Trust Score formula: 5 cases. Integration testing: 2 end-to-end scenarios (tests/test_integration.py).
Design techniques	Equivalence partitioning (valid / invalid classes), boundary-value analysis (score 39 / 40, amount above tier limit, 0 amount), negative testing, state-based testing (loan life cycle), security testing (role check, forged requests).
Environment	Windows 11, Python 3.11, Flask 3.1, SQLite 3; Chrome / Edge browser.
Tools	pytest; Flask test client (drives the real web routes); tests/run_test_cases.py records each Actual Output automatically.
Test data	Every case starts from a fresh, empty database; the data each case needs is listed in its Pre-condition column.
Entry criteria	Code of all four modules complete; application starts without errors.
Exit criteria	All documented test cases executed; every Must-have requirement traced to at least one passing test case (Section 8); no open defect.

2. Test Results Summary

Module	Test case IDs	Executed	Passed	Failed	Pass rate
User Registration & Login	TC-AUTH-01 to TC-AUTH-07	7	7	0	100%
Daily Sales Logging	TC-SALE-01 to TC-SALE-07	7	7	0	100%
Trust Score Calculation	TC-TS-01 to TC-TS-06	6	6	0	100%
Loan Application & Bank Review	TC-LOAN-01 to TC-LOAN-07	7	7	0	100%
EMI Schedule & Repayment	TC-EMI-01 to TC-EMI-07	7	7	0	100%
Total		**34**	**34**	**0**	**100%**

Automated run (pytest): 36 passed in 24.21s - the 34 documented cases above plus 2 integration tests. Executed on 02-11-2026.

How the Actual Output is obtained: tests/run_test_cases.py performs each case's steps on the running application and copies the message shown on the page into the Actual Output column; the Remark is PASS only if it matches the Expected Output exactly. Nothing in the Actual Output column is typed by hand.

Repeating a case by hand in the demo: the demo data (python seed_demo.py) already contains loans, so loan numbers such as #1 will differ; the messages are otherwise identical.

3. User Registration & Login

Test Case #:	TC-AUTH-01 to TC-AUTH-07	**Test Case Name:**	User Registration & Login
System:	VendorSetu	**Subsystem:**	Authentication (Module 1 - modules/auth.py)
Designed by:	Nancy Yadav & team	**Design Date:**	02-11-2026
Executed by:	Nancy Yadav & team	**Execution Date:**	02-11-2026
Short Description:	Verify account creation rules, login, and account lock-out after 5 wrong passwords.		

Pre-conditions
1. Application is running with an empty database.
2. User is on the Register or Log in page of VendorSetu.

TC ID	Test Scenario	Type	Pre-condition	Test Steps	Input (Test Data)	Expected Output	Actual Output	Remark
TC-AUTH-01	Register with valid details	System	No account exists for 9876543210	1. Open the Register page 2. Enter name, mobile, role, password and confirm password 3. Click 'Create account'	Name = Ramesh Kumar, Mobile = 9876543210, Role = Street Vendor, Password = ramesh123, Confirm = ramesh123	Registration successful. Please log in. -> Log in page is shown	Registration successful. Please log in. -> Log in page is shown	**PASS**
TC-AUTH-02	Register with an already registered mobile number	System	An account already exists for 9876543210	1. Open the Register page 2. Enter the same mobile number with other details 3. Click 'Create account'	Name = Ramesh Kumar, Mobile = 9876543210 (already registered), Password = ramesh123	Mobile number already registered	Mobile number already registered	**PASS**
TC-AUTH-03	Register with an invalid mobile number	System	-	1. Open the Register page 2. Enter a 5-digit mobile number 3. Click 'Create account'	Mobile = 98765 (5 digits), other fields valid	Enter a valid 10-digit mobile number	Enter a valid 10-digit mobile number	**PASS**
TC-AUTH-04	Register with a weak password	System	-	1. Open the Register page 2. Enter a 3-letter password in both password fields 3. Click 'Create account'	Password = abc, Confirm = abc, other fields valid	Password must be at least 6 characters with letters and digits	Password must be at least 6 characters with letters and digits	**PASS**
TC-AUTH-05	Register with non-matching passwords	System	-	1. Open the Register page 2. Enter different values in Password and Confirm password 3. Click 'Create account'	Password = ramesh123, Confirm = ramesh124	Passwords do not match	Passwords do not match	**PASS**
TC-AUTH-06	Log in with correct credentials	System	Ramesh is registered as a vendor	1. Open the Log in page 2. Enter mobile number and password 3. Click 'Log in'	Mobile = 9876543210, Password = ramesh123	Welcome, Ramesh Kumar! -> Vendor dashboard is shown	Welcome, Ramesh Kumar! -> Vendor dashboard is shown	**PASS**
TC-AUTH-07	Account lock after 5 wrong passwords	System	Ramesh is registered as a vendor	1. Log in with a wrong password 5 times 2. Then log in with the correct password	Mobile = 9876543210; Password = wrong1 (x5), then ramesh123	Attempts 1-4: Invalid mobile number or password | Attempt 5: Account locked after 5 failed attempts. Contact support | Attempt 6 (correct password): Account locked after 5 failed attempts. Contact support	Attempts 1-4: Invalid mobile number or password | Attempt 5: Account locked after 5 failed attempts. Contact support | Attempt 6 (correct password): Account locked after 5 failed attempts. Contact support	**PASS**

Post-conditions
1. A valid registration stores exactly one user with a hashed password; rejected input stores nothing.
2. After 5 consecutive wrong passwords the account stays locked even for the correct password.

4. Daily Sales Logging

Test Case #:	TC-SALE-01 to TC-SALE-07	**Test Case Name:**	Daily Sales Logging
System:	VendorSetu	**Subsystem:**	Vendor Profile & Sales (Module 2 - modules/vendor.py)
Designed by:	Nancy Yadav & team	**Design Date:**	02-11-2026
Executed by:	Nancy Yadav & team	**Execution Date:**	02-11-2026
Short Description:	Verify that a vendor can log one valid sales entry per day and that invalid entries are rejected.		

Pre-conditions
1. Vendor 'Ramesh Kumar' (9876543210 / ramesh123) is registered and logged in.
2. His business profile is saved (unless the case says otherwise).

TC ID	Test Scenario	Type	Pre-condition	Test Steps	Input (Test Data)	Expected Output	Actual Output	Remark
TC-SALE-01	Log a valid daily sale	System	No sale logged today	1. Log in as Ramesh (9876543210 / ramesh123) 2. In 'Daily sales', enter the date, amount and payment mode 3. Click 'Log sale'	Date = 02-11-2026 (today), Amount = 1500, Mode = UPI	Sale logged. Your trust score has been updated.	Sale logged. Your trust score has been updated.	**PASS**
TC-SALE-02	Log a sale with zero amount	System	-	1. Log in as Ramesh (9876543210 / ramesh123) 2. In 'Daily sales', enter the date, amount and payment mode 3. Click 'Log sale'	Date = 02-11-2026, Amount = 0, Mode = Cash	Sale amount must be greater than 0	Sale amount must be greater than 0	**PASS**
TC-SALE-03	Log a sale for a future date	System	-	1. Log in as Ramesh (9876543210 / ramesh123) 2. In 'Daily sales', enter the date, amount and payment mode 3. Click 'Log sale'	Date = 04-11-2026 (2 days ahead), Amount = 1200, Mode = Cash	Sale date cannot be in the future	Sale date cannot be in the future	**PASS**
TC-SALE-04	Log a second sale for the same date	System	A sale for 01-11-2026 is already logged	1. Log in as Ramesh (9876543210 / ramesh123) 2. In 'Daily sales', enter the date, amount and payment mode 3. Click 'Log sale'	Date = 01-11-2026, Amount = 900, Mode = Cash	Sales already logged for this date	Sales already logged for this date	**PASS**
TC-SALE-05	Log a sale above the daily limit	System	-	1. Log in as Ramesh (9876543210 / ramesh123) 2. In 'Daily sales', enter the date, amount and payment mode 3. Click 'Log sale'	Date = 02-11-2026, Amount = 150000, Mode = UPI	Sale amount cannot exceed Rs 1,00,000 per day	Sale amount cannot exceed Rs 1,00,000 per day	**PASS**
TC-SALE-06	Log a sale before saving the business profile	System	Ramesh has NOT saved his business profile	1. Log in as Ramesh (9876543210 / ramesh123) 2. In 'Daily sales', enter the date, amount and payment mode 3. Click 'Log sale'	Date = 02-11-2026, Amount = 1500, Mode = UPI	Complete your vendor profile first	Complete your vendor profile first	**PASS**
TC-SALE-07	30-day summary on the dashboard	System	No sales logged yet	1. Log three sales (one per day) 2. Open the dashboard and read the summary cards	31-10-2026: 1000, 01-11-2026: 2000, 02-11-2026: 3000	30-day sales Rs 6,000; active days 3/30; average Rs 2,000 per active day	30-day sales Rs 6,000; active days 3/30; average Rs 2,000 per active day	**PASS**

Post-conditions
1. A valid entry is saved and counted in the 30-day summary; a rejected entry changes nothing.

5. Trust Score Calculation

Test Case #:	TC-TS-01 to TC-TS-06	**Test Case Name:**	Trust Score Calculation
System:	VendorSetu	**Subsystem:**	Alt-Credit Scoring Engine (Module 3 - modules/credit_score.py)
Designed by:	Nancy Yadav & team	**Design Date:**	02-11-2026
Executed by:	Nancy Yadav & team	**Execution Date:**	02-11-2026
Short Description:	Verify the 0-100 Trust Score formula, the risk band and the loan-eligibility cut-off of 40.		

Pre-conditions
1. Formula: Sales (active days / 30 x 40) + Repayment (on-time / due x 40, or 20 if no EMI is due yet) + Peer vouching (min(vouches, 3) / 3 x 10) + Tenure (min(years, 5) / 5 x 10), rounded.
2. Bands: 70-100 LOW RISK, 40-69 MEDIUM RISK, 0-39 HIGH RISK.

TC ID	Test Scenario	Type	Pre-condition	Test Steps	Input (Test Data)	Expected Output	Actual Output	Remark
TC-TS-01	Ideal vendor - maximum observable score	Unit	No EMI due yet	1. Call compute_trust_score() with the input values	Active days = 30/30, EMIs due = 0, Vouches = 3, Years = 5	40 + 20 + 10 + 10 = 80 -> LOW RISK, eligible	40 + 20 + 10 + 10 = 80 -> LOW RISK, eligible	**PASS**
TC-TS-02	Boundary value - score exactly 40	Unit	No EMI due yet	1. Call compute_trust_score() with the input values	Active days = 15/30, EMIs due = 0, Vouches = 0, Years = 0	20 + 20 + 0 + 0 = 40 -> MEDIUM RISK, eligible	20 + 20 + 0 + 0 = 40 -> MEDIUM RISK, eligible	**PASS**
TC-TS-03	Boundary value - score just below 40	Unit	No EMI due yet	1. Call compute_trust_score() with the input values	Active days = 14/30, EMIs due = 0, Vouches = 0, Years = 0	18.7 + 20 + 0 + 0 = 39 -> HIGH RISK, not eligible	18.7 + 20 + 0 + 0 = 39 -> HIGH RISK, not eligible	**PASS**
TC-TS-04	Vendor with a partial repayment record	Unit	-	1. Call compute_trust_score() with the input values	Active days = 24/30, On-time EMIs = 6 of 12 due, Vouches = 1, Years = 2	32 + 20 + 3.3 + 4 = 59 -> MEDIUM RISK, eligible	32 + 20 + 3.3 + 4 = 59 -> MEDIUM RISK, eligible	**PASS**
TC-TS-05	Invalid input - active days exceed the window	Unit	-	1. Call compute_trust_score() with the input values	Active days = 35, Window = 30	Active days cannot exceed the observation window	Active days cannot exceed the observation window	**PASS**
TC-TS-06	New vendor sees the score on the dashboard	System	Vendor 'Sunita Devi' (9812345670 / sunita123) has a profile with 0 years of vending and no sales	1. Log in as Sunita 2. Read the Trust Score card	Mobile = 9812345670, Password = sunita123	Dashboard shows Trust Score 20/100, HIGH RISK	Dashboard shows Trust Score 20/100, HIGH RISK	**PASS**

Post-conditions
1. Score, band and eligibility are returned / displayed; no stored data is modified.

6. Loan Application & Bank Review

Test Case #:	TC-LOAN-01 to TC-LOAN-07	**Test Case Name:**	Loan Application & Bank Review
System:	VendorSetu	**Subsystem:**	Loan Management (Module 4 - modules/loan.py)
Designed by:	Nancy Yadav & team	**Design Date:**	02-11-2026
Executed by:	Nancy Yadav & team	**Execution Date:**	02-11-2026
Short Description:	Verify progressive-tier loan application rules and the bank officer's approve / reject / disburse actions.		

Pre-conditions

1. Vendor 'Ramesh Kumar' (9876543210 / ramesh123): profile with 5 years of vending and a sale logged on each of the last 30 days, so his Trust Score is 40 + 20 + 0 + 10 = 70.
2. Bank officer 'Anil Sharma' (9123456780 / bank123) is registered.

TC ID	Test Scenario	Type	Pre-condition	Test Steps	Input (Test Data)	Expected Output	Actual Output	Remark
TC-LOAN-01	Eligible vendor applies for a Tier-1 loan	System	Ramesh has no loan	1. Log in as Ramesh (9876543210 / ramesh123) 2. In 'Micro-credit', enter the amount 3. Click 'Apply for loan'	Amount = 15000 (Trust Score 70, Tier 1 limit 15,000)	Loan application #1 submitted. Status: PENDING.	Loan application #1 submitted. Status: PENDING.	**PASS**
TC-LOAN-02	Amount above the tier limit	System	Ramesh has no loan	1. Log in as Ramesh (9876543210 / ramesh123) 2. Enter an amount above Rs 15,000 3. Click 'Apply for loan'	Amount = 20000	Amount exceeds Tier 1 limit of Rs 15,000	Amount exceeds Tier 1 limit of Rs 15,000	**PASS**
TC-LOAN-03	Vendor with a low Trust Score applies	System	Vendor 'Sunita Devi' (9812345670 / sunita123) has a profile with 0 years of vending and no sales (score 20)	1. Log in as Sunita 2. Enter the amount 3. Click 'Apply for loan'	Amount = 10000	Trust score 20 is below 40. Not eligible	Trust score 20 is below 40. Not eligible	**PASS**
TC-LOAN-04	Second application while one is pending	System	Ramesh's application #1 for Rs 15,000 is PENDING	1. Log in as Ramesh (9876543210 / ramesh123) 2. Enter the amount 3. Click 'Apply for loan'	Amount = 5000	You already have an active loan	You already have an active loan	**PASS**
TC-LOAN-05	Officer rejects without remarks	System	Ramesh's application #1 is PENDING	1. Log in as bank officer (9123456780 / bank123) 2. Leave the Remarks box empty 3. Click 'Reject' on application #1	Remarks = (empty), Decision = Reject	Remarks are required when rejecting	Remarks are required when rejecting	**PASS**
TC-LOAN-06	Vendor tries to open the bank officer page	System	-	1. Log in as Ramesh (9876543210 / ramesh123) 2. Type the officer page address in the browser	URL = /officer	You are not allowed to open that page.	You are not allowed to open that page.	**PASS**
TC-LOAN-07	Officer approves and disburses	System	Ramesh's application #1 is PENDING	1. Log in as bank officer (9123456780 / bank123) 2. Click 'Approve' on application #1 3. Click 'Disburse' on application #1	Decision = Approve, then Disburse	Loan #1 APPROVED. | Loan #1 disbursed. 12 EMIs of Rs 1332.73 scheduled.	Loan #1 APPROVED. | Loan #1 disbursed. 12 EMIs of Rs 1332.73 scheduled.	**PASS**

Post-conditions

1. An accepted application is stored as PENDING; approval and disbursal change its status and create the EMI schedule; rejected requests change nothing.

7. EMI Schedule & Repayment

Test Case #:	TC-EMI-01 to TC-EMI-07	**Test Case Name:**	EMI Schedule & Repayment
System:	VendorSetu	**Subsystem:**	Loan Management - Repayment (Module 4 - modules/loan.py)
Designed by:	Nancy Yadav & team	**Design Date:**	02-11-2026
Executed by:	Nancy Yadav & team	**Execution Date:**	02-11-2026
Short Description:	Verify the EMI schedule, the repayment rules, loan closure (next tier unlocked) and the default check.		

Pre-conditions
1. Ramesh's Tier-1 loan #1 of Rs 15,000 at 12% p.a. for 12 months has been approved and disbursed today by the bank officer (unless the case says otherwise).
2. Hand calculation: r = 12% / 12 = 0.01, EMI = 15000 x 0.01 x 1.01^12 / (1.01^12 - 1) = Rs 1,332.73.

TC ID	Test Scenario	Type	Pre-condition	Test Steps	Input (Test Data)	Expected Output	Actual Output	Remark
TC-EMI-01	View the EMI schedule	System	-	1. Log in as Ramesh (9876543210 / ramesh123) 2. Click 'View' on loan #1 3. Read the EMI schedule table	Loan #1: P = 15000, r = 12% p.a., n = 12, disbursed 02-11-2026	12 installments; EMI #1 Rs 1,332.73 due 2026-12-02; EMI #12 due 2027-11-02	12 installments; EMI #1 Rs 1,332.73 due 2026-12-02; EMI #12 due 2027-11-02	**PASS**
TC-EMI-02	Pay EMI #1 before its due date	System	No EMI paid yet	1. Log in as Ramesh (9876543210 / ramesh123) 2. Open loan #1 3. Click 'Pay' on installment #1 (amount pre-filled)	Installment = 1, Amount = 1332.73, paid on 02-11-2026	EMI #1 paid on time.	EMI #1 paid on time.	**PASS**
TC-EMI-03	Pay less than the EMI amount	System	No EMI paid yet	1. Log in as Ramesh (9876543210 / ramesh123) 2. Open loan #1 3. Change the amount to 1000 4. Click 'Pay' on installment #1	Installment = 1, Amount = 1000	Amount is less than EMI due of Rs 1332.73	Amount is less than EMI due of Rs 1332.73	**PASS**
TC-EMI-04	Skip an installment (forged request)	System	EMI #1 is paid; #2 is due	1. Log in as Ramesh (9876543210 / ramesh123) 2. Pay EMI #1 3. Send a payment request for installment #3 directly (the page only offers #2)	POST /vendor/loan/1/pay/3, Amount = 1332.73	Pay earlier installment #2 first	Pay earlier installment #2 first	**PASS**
TC-EMI-05	Pay the same installment twice (forged request)	System	EMI #1 is already paid	1. Log in as Ramesh (9876543210 / ramesh123) 2. Pay EMI #1 3. Send a payment request for installment #1 again	POST /vendor/loan/1/pay/1, Amount = 1332.73	Installment already paid	Installment already paid	**PASS**
TC-EMI-06	Repay all 12 EMIs (progressive lending)	System	No EMI paid yet	1. Log in as Ramesh (9876543210 / ramesh123) 2. Pay installments 1 to 12 in order 3. Open the dashboard	12 payments, each equal to the EMI shown	EMI #12 paid on time. Loan fully repaid - you are now eligible for the next tier! -> Dashboard shows eligible limit Rs 25,000 (Tier 2)	EMI #12 paid on time. Loan fully repaid - you are now eligible for the next tier! -> Dashboard shows eligible limit Rs 25,000 (Tier 2)	**PASS**
TC-EMI-07	Default check marks a 90+ days overdue loan	System	Loan #1 was disbursed on 05-06-2026 (150 days ago) and no EMI has been paid	1. Log in as bank officer (9123456780 / bank123) 2. Click 'Run default check (90+ days overdue)'	(button click, no data)	Default check complete: 1 loan(s) marked DEFAULTED.	Default check complete: 1 loan(s) marked DEFAULTED.	**PASS**

Post-conditions
1. Paid installments are marked PAID (on time / late); when all 12 are paid the loan is CLOSED and Tier 2 (Rs 25,000) is unlocked.

8. Requirements Traceability Matrix

Each requirement from the SRS that the implemented modules cover, and the test cases that verify it.

Requirement	Verified by test case(s)	Status
FR-AUTH-1	TC-AUTH-01, TC-AUTH-05	Covered - PASS
FR-AUTH-2	TC-AUTH-02, TC-AUTH-03	Covered - PASS
FR-AUTH-3	TC-AUTH-04	Covered - PASS
FR-AUTH-5	TC-AUTH-06	Covered - PASS
FR-AUTH-6	TC-AUTH-07	Covered - PASS
FR-AUTH-7	TC-LOAN-06	Covered - PASS
FR-VEN-3	TC-SALE-01	Covered - PASS
FR-VEN-4	TC-SALE-02, TC-SALE-05	Covered - PASS
FR-VEN-5	TC-SALE-03, TC-SALE-04	Covered - PASS
FR-VEN-6	TC-SALE-06	Covered - PASS
FR-VEN-7	TC-SALE-07	Covered - PASS
FR-TS-1	TC-TS-05	Covered - PASS
FR-TS-2	TC-TS-01, TC-TS-02, TC-TS-03, TC-TS-04	Covered - PASS
FR-TS-3	TC-TS-01, TC-TS-06	Covered - PASS
FR-TS-4	TC-TS-01, TC-TS-02, TC-TS-03, TC-TS-04, TC-TS-06	Covered - PASS
FR-TS-5	TC-TS-02, TC-TS-03, TC-LOAN-03	Covered - PASS
FR-TS-6	TC-TS-06	Covered - PASS
FR-LN-1	TC-LOAN-01	Covered - PASS
FR-LN-2	TC-LOAN-02	Covered - PASS
FR-LN-3	TC-EMI-06	Covered - PASS
FR-LN-4	TC-LOAN-02	Covered - PASS
FR-LN-5	TC-LOAN-04	Covered - PASS
FR-LN-6	TC-LOAN-01	Covered - PASS
FR-BNK-2	TC-LOAN-07	Covered - PASS
FR-BNK-3	TC-LOAN-05	Covered - PASS
FR-BNK-4	TC-LOAN-06	Covered - PASS
FR-BNK-5	TC-LOAN-07	Covered - PASS
FR-BNK-6	TC-LOAN-07, TC-EMI-01	Covered - PASS
FR-BNK-7	TC-EMI-07	Covered - PASS
FR-EMI-1	TC-EMI-01	Covered - PASS
FR-EMI-2	TC-EMI-04	Covered - PASS
FR-EMI-3	TC-EMI-03	Covered - PASS
FR-EMI-4	TC-EMI-05	Covered - PASS
FR-EMI-5	TC-EMI-02	Covered - PASS
FR-EMI-6	TC-EMI-06	Covered - PASS
BR-1	TC-LOAN-06	Covered - PASS
BR-2	TC-LOAN-04	Covered - PASS
BR-4	TC-LOAN-03	Covered - PASS
BR-5	TC-LOAN-05	Covered - PASS

Requirement	Verified by test case(s)	Status
BR-6	TC-EMI-07	Covered - PASS
NFR-SEC-2	TC-AUTH-07	Covered - PASS

9. How to Re-execute the Tests

```
cd SE_Project
python tests/run_test_cases.py      (runs every case above, prints PASS/FAIL, saves actual outputs)
python -m pytest tests -v           (same cases + 2 integration tests)
python docs/build_test_cases.py     (regenerates this document from the latest run)
```